P1 PUMP / PRODUCT INFORMATION

P1 Axial Piston Pump

The P1's 0.8 and 1.2 cc/rev (0.0488 and 0.0732 in³/rev) displacement options support small water hydraulic circuits. Check delivered flow at the required speed and pressure, then confirm inlet conditions, water specification and integration space.

The P1 is not an oil hydraulic pump with water bolted on. Every internal is purpose-built for water as the working fluid. CF30-PEEK (carbon fibre reinforced PEEK) components resist corrosion and wear in fresh or salt water environments without the lubrication oil-based systems depend on. The result is a pump that is genuinely oil-free from drive shaft to port face.



P1 axial piston water hydraulic pump, 316L stainless steel construction

At a glance

Parameter / value	Parameter / value
Displacement 0.8 / 1.2 cc/rev (0.073 in³/rev (1.2 cm³/rev))	Max Pressure 160 bar (2,321 psi (160 bar (2,321 psi)))
Max Flow 2.1 L/min (0.55 US gpm (2.08 L/min (0.55 US gpm)))	Speed Range 500-1750 rpm
Working Fluid Tap / Sea Water	

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Parameter	Options	Notes
Displacement	0.8 cc/rev (0.0488 in³/rev) (0.049 in³/rev) 1.2 cc/rev (0.0732 in³/rev) (0.073 in³/rev)	1.2cc (0.0732 in³) recommended for most HPU and actuator applications
Working Fluid	Tap Water Seawater Treated Process Water	Specify fluid chemistry for seal and port material selection
Port Standard	BSPP (G) NPT Metric	BSPP standard; specify NPT or metric for North American OEM integration
Shaft Configuration	Keyed Shaft Splined Flange Mount	Keyed shaft fits standard IEC motor couplings direct
Pressure Rating	Up to 160 bar (2,321 psi)	Single rated range; specify operating point for system sizing support
Supply as	Bare Pump HPU Package Integrated Assembly	HPU package includes motor, reservoir, filter, manifold and pressure gauge
Certifications	Confirm applicability and supplied documentation: CE RoHS Material Test Reports	Confirm applicability and supplied documentation: 316L MTRs available on request for compliance documentation

Configuration and ordering data

Displacement	0.8 cc/rev (0.0488 in³/rev) (0.049 in³/rev) / 1.2 cc/rev (0.0732 in³/rev) (0.073 in³/rev)
Max Operating Pressure	160 bar (2,320 psi)
Max Flow at 1750 rpm	1.4 L/min (0.37 US gpm) (0.8cc (0.0488 in³)) / 2.1 L/min (0.55 US gpm) (1.2cc (0.0732 in³))
Speed Range	500 to 1,750 rpm
Peak Overall Efficiency	77% at 40 bar (580 psi), 1500 rpm (1.2cc (0.0732 in³))
Max Input Torque (1.2cc (0.0732 in³))	3.2 Nm (2.36 lbf·ft) at 160 bar (2,321 psi)
Max Input Power (1.2cc (0.0732 in³))	0.59 kW (0.79 hp) at 160 bar (2,321 psi), 1750 rpm

Configuration and ordering data

Pump Type	External Gear Pump
Housing Material	316L Stainless Steel
Internal Components	CF30-PEEK (Carbon-Reinforced PEEK)
Seals	EPDM / PTFE (water-compatible)
Working Fluid	Tap water, seawater, process water
Fluid Temperature	5 to 60 degrees C
Filtration Requirement	10 micron (394 µin) (0.394 mil) absolute (inlet)

Configuration and ordering data

Port Standard	BSPP / NPT / Metric (specify)
Inlet Port Size	G1/4" (102 mm (4.02 in)) (standard)
Outlet Port Size	G1/4" (102 mm (4.02 in)) (standard)
Shaft Type	Keyed / Splined / Flange (specify)
Rotation Direction	Clockwise (standard) / CCW (specify)
Mounting Pattern	SAE A 2-bolt (standard)

Configuration and ordering data

Certifications	Confirm applicability and supplied documentation: CE marked, RoHS compliant
Material Reports	316L MTRs available on request
Environmental	No hydraulic oil; Fisheries Act compatible
Origin	Designed and assembled in Canada
Warranty	12 months from commissioning
Lead Time	Contact WaterQuest for current availability